



Politecnico
di Torino

Algorithms and Programming

Iniziato	martedì, 21 dicembre 2021, 17:09
Stato	Completato
Terminato	martedì, 21 dicembre 2021, 17:09
Tempo impiegato	15 secondi
Valutazione	0,00 su un massimo di 9,00 (0%)

Domanda 1

Risposta non data

Punteggio max.:
2,00

The main program calls function f as f(3, 5). In turn function f calls function g.

```
void f (int n, int m) {  
    printf ("F");  
    if (n<=0)  
        return;  
    if (n%2==0) {  
        g(n,m-1);  
    } else {  
        f(n-1,m);  
    }  
    return;  
}  
  
void g (int n, int m) {  
    printf ("G");  
    if (m<=0)  
        return;  
    if (m%2==0) {  
        g(n,m-1);  
    } else {  
        f(n-1,m);  
    }  
    return;  
}
```

Following the recursion tree of f and g, indicate the number of times the character 'F' and the character 'G' are displayed on standard output.

Please, report your response as a pair of integer values (the first one for 'F', the second one for 'G'). No other symbols must be included in the response. For example, if the function displays "FGFGFGGG", you must reply: 3 5

Risposta:



La risposta corretta è : 4 2

Domanda 2

Risposta non data

Punteggio max.:
3,00

A string has an undefined length and it includes all possible characters.

Write a non-recursive function that receives such a string as a parameter and it computes the length of the longest sub-sequence made only of small alphabetic characters and the length of the longest sub-sequence made only of decimal digits.

The function prototype is the following:

```
void substring (char *str, int *letter, int *digit);
```

where **str** is the string received by the function, and **letter** and **digit** are the two integers, passed by reference, which represent the desired return values.

For example, if the input string is "This is 1 string inclUding diGIts: 12345 678 9", the longest sequence of consecutive small characters is "string" and the longest sequence of digits is "12345" (please, notice spaces are neither alphabetic characters nor digits). Thus, the program must assign and return variable **letter** equal to 6 and **digit** equal to 5.

Please, report the entire C code of the function. Briefly explain (in plain English language, within a C comment) the logic of the code.

Domanda 3

Risposta non data

Punteggio max.:

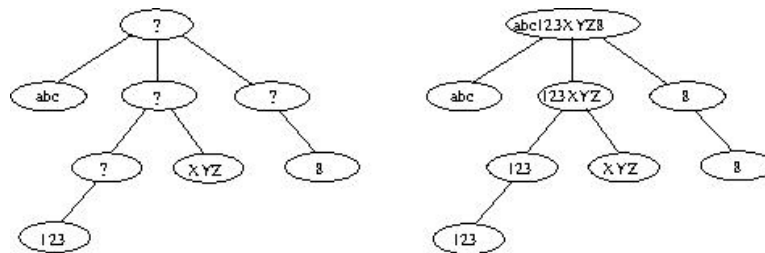
4,00

A n-ary tree is organized adopting the following C data structure:

```
typedef struct e_s e_t;
struct e_s {
    char *str;
    int n;
    e_t **children;
};
```

where **str** is the identifier of the node of the tree, **n** is the number of children of the node, and **children** is the array of pointers referring to these children.

At the very beginning, the field **str** is defined only for the leaves of the tree, as represented by the leftmost tree of the following picture. For all internal nodes, **str** is not NULL but it is undefined (i.e., it points to an undefined, dynamically allocated, and useless string).



Write the function

```
<type> tree_complete (e_t *root);
```

which receives the tree rooted at **root** and it recursively completes the tree by assigning to each field **str** (but the ones on the leaves that have to remain untouched) a string which is the concatenation of the strings stored in the node's children. The result of this operation for the leftmost tree of the picture should deliver the rightmost tree of the same picture. The **type** of the function **tree_complete**, i.e., **<type>**, can be selected by the candidate.

Please, report the entire C code of the function. Allocate and free all required dynamic fields avoiding memory leaks. Briefly explain (in plain English language, within a C comment) the logic of the code.

